Notes: Harvey & Whaley (JF, 1991)

S&P 100 Index Option Volatility

Contents

1 Big picture

- **Question.** How should researchers estimate implied market volatility from S&P 100 index options, and what biases arise from common simplifications?
- **Setting.** The paper studies S&P 100 index option transactions from August 1, 1988 to July 31, 1989.
- **Main object.** The object of interest is the time series of implied volatility inferred from S&P 100 index call and put options.
- **Core critique.** Prior studies often use closing option prices, assume European exercise, and approximate dividends with a constant yield. Harvey and Whaley argue that these shortcuts can create misleading volatility dynamics.
- **Valuation correction.** The paper uses an American-style option pricing approximation and the exact discrete cash dividend series for the S&P 100 index.
- **Nonsynchronous price problem.** The stock market closes at 3:00 P.M., while the S&P 100 option market closes at 3:15 P.M. Using the option close with the index close creates artificial volatility movements.
- **Bid/ask effect.** A single option transaction can occur at the bid or ask. This bid/ask bounce induces negative serial correlation in implied volatility changes.
- **Preferred estimator.** The paper uses all option transactions in the 2:55–3:05 P.M. window and estimates volatility by nonlinear least squares.
- **Infrequent trading test.** The authors also pool calls and puts and estimate both the implied volatility and the implied index level. Results are almost unchanged, so stale index-stock prices are not the main problem.
- **Main result.** Closing-price implied volatilities show strong artificial negative serial correlation and negative call-put cross-correlations. Using transaction-window estimators greatly improves the diagnostics.
- **Takeaway.** Implied volatility is economically useful only if the option-pricing model, dividend treatment, and timing of observed prices are handled carefully.

2 Data and measurement

2.1 The S&P 100 index option

- The S&P 100 index option, OEX, trades on the Chicago Board Options Exchange.
- It is an American-style option, not a European-style option.
- The underlying index pays a sequence of discrete dividends, not a constant dividend yield.
- The paper uses at-the-money call and put options with the shortest maturity, subject to having at least 15 days to expiration.

Interpretation: At-the-money options are used because they are most sensitive to volatility. This makes their prices the most informative for implied-volatility estimation.

2.2 Market inputs

For each trading day, the implied-volatility calculation uses:

- the S&P 100 index level;

- the Treasury bill rate matched to option maturity, or the 30-day rate when that is the longest available maturity;
- the actual discrete dividends paid during the option's life;
- option transaction prices.

Interpretation: The paper is not merely changing the econometric estimator. It also fixes the economic valuation environment by using American exercise and discrete dividends.

2.3 Vega and the at-the-money choice

For a European option, option vega is

$$\frac{\partial v}{\partial \sigma} = Sn(d_1)\sqrt{T}, \tag{1}$$

where

$$d_1 = \frac{\ln(S/X) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}}. \tag{2}$$

- S is the index level.
- X is the strike price.
- r is the riskless interest rate.
- σ is volatility.
- T is time to expiration.
- $n(\cdot)$ is the standard normal density.

Interpretation: Vega is largest near the money because the normal density is largest near zero. This is why the paper uses at-the-money options rather than deeply in- or out-of-the-money contracts.

2.4 Three main volatility estimators

1. **Closing-price estimator.** Uses the 3:15 P.M. option closing price and the 3:00 P.M. stock-index close.
2. **3:00 P.M. transaction estimator.** Uses the last option transaction before 3:00 P.M. and the 3:00 P.M. stock-index close.
3. **2:55–3:05 P.M. nonlinear least squares estimator.** Uses all option transactions in a ten-minute window around the stock-market close.

Interpretation: The first estimator is exposed to nonsynchronous closing prices. The second fixes the closing-time mismatch but still uses a single transaction. The third uses many transactions to reduce bid/ask bounce.

2.5 Nonlinear least squares estimator

Let P_j^{mkt} be the observed price of option transaction j in the 2:55–3:05 P.M. window. Let $P_j^{model}(\sigma; S, r, D)$ be the American-style model price with discrete dividends D. The estimator is

$$\widehat{\sigma} = \arg\min_{\sigma} \sum_{j=1}^{N} \left[P_j^{mkt} - P_j^{model}(\sigma; S, r, D)\right]^2. \tag{3}$$

For the infrequent-trading test, the paper also estimates

$$(\widehat{\sigma}, \widehat{S}) = \arg\min_{\sigma,S} \sum_{j=1}^{N} \left[P_j^{mkt} - P_j^{model}(\sigma; S, r, D)\right]^2. \tag{4}$$

Interpretation: Estimating S along with σ removes reliance on the observed index level and tests whether stale component-stock prices are driving the results.

3 Models and empirical specifications

3.1 Nonsynchronous closing-price mechanism

The stock market closes at 3:00 P.M., but the OEX option market closes at 3:15 P.M. If new information arrives after 3:00 P.M., option prices update but the index close does not. Therefore:

- good news after 3:00 P.M. raises call prices and lowers put prices relative to the stale index level;
- bad news after 3:00 P.M. lowers call prices and raises put prices relative to the stale index level;
- the next day the stock index catches up, so the implied-volatility error reverses.

Interpretation: This creates spurious negative serial correlation in daily changes in implied volatility and can also create negative call-put cross-correlation.

3.2 Bid/ask price effect

A single option transaction can occur near the bid or near the ask. If today's transaction occurs at the ask and tomorrow's at the bid, the implied-volatility change looks negative even when true volatility is unchanged.

Interpretation: Bid/ask bounce can produce negative first-order serial correlation in volatility changes. Using many transactions within a window averages out bid and ask observations.

3.3 Call-put consistency test

Calls and puts on the same index should imply changes in the same underlying volatility. A reliable estimator should therefore produce positive contemporaneous correlation between call and put implied-volatility changes.

Interpretation: If calls and puts imply negatively correlated volatility changes, the problem is probably measurement error rather than an economically meaningful volatility pattern.

3.4 Infrequent-trading test

If the observed index level is stale because some constituent stocks do not trade near the close, call and put implied volatilities should move in opposite directions. The paper tests this in two ways:

1. pool call and put transactions in the 2:55–3:05 P.M. window and estimate one volatility;
2. estimate both implied volatility and the implied index level from pooled call and put transactions.

Interpretation: The results barely change, indicating that infrequent trading of S&P 100 stocks near the close is not the main source of volatility-measurement error.

4 Identification and empirical design

4.1 What identifies the measurement biases

- The same option contracts are used under different observation rules.
- Comparing closing prices with 3:00 P.M. prices isolates nonsynchronous closing effects.
- Comparing one-transaction estimates with multi-transaction estimates isolates bid/ask bounce.
- Comparing observed-index and implied-index estimates tests stale index-price effects.

Interpretation: This is a measurement-design paper. Identification comes from changing the estimator while keeping the underlying market and sample fixed.

4.2 Why the 2:55–3:05 P.M. window matters

- It is close to the stock-market close.
- It includes transactions just before and just after 3:00 P.M., allowing for reporting delays.
- It provides dozens of transactions per day, enough to reduce bid/ask noise.

Interpretation: The window balances simultaneity with the need for enough option transactions to estimate volatility precisely.

4.3 What the paper cannot fully prove

- The sample covers only one year, August 1988 through July 1989.
- The paper studies near-close implied volatility, not intraday volatility dynamics throughout the entire day.
- It focuses on at-the-money near-maturity options.
- The remaining negative serial correlation may reflect true mean reversion in volatility or residual measurement error.

Interpretation: The paper's strongest claim is that common implied-volatility estimators mechanically induce patterns that researchers may mistakenly interpret as market inefficiency or volatility dynamics.

5 Tables and figures

The paper has no plotted figures. The central visual evidence is in two tables, directly cropped from the original paper.

5.1 Table I: Summary statistics for implied volatility estimates

Read:

- The mean call implied volatility is 15.88% using closing prices, 15.89% using the last transaction before 3:00 P.M., and 15.99% using 2:55–3:05 P.M. transactions.
- The mean put implied volatility is 16.62%, 16.70%, and 16.58% under the same three estimators.
- The pooled call-put estimator gives 16.24% using the observed index level and 16.30% when the index level is jointly implied.
- Standard deviations fall when multiple transactions are used, consistent with a more precise estimator.

- The average ten-minute window contains 28.21 call transactions, 26.83 put transactions, and 55.04 pooled transactions.

Economic message: The average level of volatility is not the main problem. The important differences appear in the time-series diagnostics of volatility changes.

Summary Statistics for Implied Volatility Estimates[a] from S&P 100 Index Call and Put Option Prices August 1, 1988–July 31, 1989

The implied volatilities are based on an American option pricing approximation using discrete cash dividend payments on the underlying index. The volatility is computed for the closest at-the-money call option and put option each day during the sample period. The riskless rate of interest is the rate on the T-bill that matures nearest after the option expires and that has at least 30 days to maturity. The first implied volatility uses the closing option prices, 3:15 P.M. (CST), and the closing index level, 3:00 P.M. (CST). The second implied volatility uses the last option transaction price prior to 3:00 P.M. (CST) and the closing index level. The third implied volatility is based on a nonlinear least squares regression of option transaction prices during the 10-minute interval 2:55–3:05 P.M. (CST) on model prices using the contemporaneous index levels. The implied volatility measures for calls and puts are based on nonlinear least squares regressions of (a) call and put option transaction prices during the 10-minute interval 2:55–3:05 P.M. (CST) on model prices using the contemporaneous index levels and (b) call and put option transaction prices during the 10-minute interval 2:55–3:05 P.M. (CST) on model prices, estimating simultaneously the implied index levels.

	Call options			Put options			Call and put options 2:55–3:05 P.M. prices	
	Closing prices	3:00 P.M. prices	2:55–3:05 P.M. prices	Closing prices	3:00 P.M. prices	2:55–3:05 P.M. prices	Observed index	Implied index
Estimator i	(1)	(2)	(3)	(1)	(2)	(3)		
Mean σ_i	15.88	15.89	15.99	16.62	16.70	16.58	16.24	16.30
Std. dev. σ_i	2.53	2.44	2.18	2.44	2.33	2.30	2.16	2.13
Average no. of observations			28.21			26.83	55.04	55.04
Mean $(\sigma_i - \sigma_3)$	−0.11	−0.09		0.04	0.13			
Std. dev. $(\sigma_i - \sigma_3)$	1.19	0.77		0.78	0.58			

[a] $n = 253$.

5.2 Table II: Diagnostics of implied-volatility changes

Read:

- Using closing prices, first-order autocorrelation in volatility changes is strongly negative: −0.4414 for calls and −0.3934 for puts.
- Using the last transaction before 3:00 P.M. reduces this to −0.3274 for calls and −0.3321 for puts.
- Using 2:55–3:05 P.M. transactions reduces it further to −0.2810 for calls and −0.3075 for puts.
- Pooling calls and puts gives first-order autocorrelations of −0.2562 with the observed index and −0.2460 with the implied index.
- Closing-price call-put cross-correlation at lag zero is negative, −0.1858.
- The 2:55–3:05 P.M. estimator changes this to a positive contemporaneous cross-correlation of 0.2738.

Economic message: The diagnostics show that nonsynchronous prices and bid/ask bounce create spurious volatility patterns. A transaction-window estimator produces more plausible call-put behavior.

Table II

Diagnostics of Changes in Implied Volatility Estimates[a] from S&P 100 Index Call and Put Option Prices August 1, 1988–July 31, 1989

The implied volatilities are based on an American option pricing approximation using discrete cash dividend payments on the underlying index. The volatility is computed for the closest at-the-money call option and put option each day during the sample period. The riskless rate of interest is the rate on the T-bill that matures nearest after the option expires and that has at least 30 days to maturity. The first implied volatility uses the closing option prices, 3:15 P.M. (CST), and the closing index level, 3:00 P.M. (CST). The second implied volatility uses the last option transaction price prior to 3:00 P.M. (CST) and the closing index level. The third implied volatility is based on a nonlinear least squares regression of option transaction prices during the 10-minute interval 2:55–3:05 P.M. (CST) on model prices using the contemporaneous index levels. The implied volatility measures for call and puts are based on nonlinear least squares regressions of (a) call and put option transaction prices during the 10-minute interval 2:55–3:05 P.M. (CST) on model prices using the contemporaneous index levels and (b) call and put option transaction prices during the 10-minute interval 2:55–3:05 P.M. (CST) on model prices, estimating simultaneously the implied index levels.

Panel A. Serial Correlation of Implied Volatility Changes

	Call options			Put options			Call and put options 2:55–3:05 P.M.	
Lag k	Closing prices	3:00 P.M. prices	2:55–3:05 P.M. prices	Closing prices	3:00 P.M. prices	2:55–3:05 P.M. prices	Observed index	Implied index
1	−0.4414	−0.3274	−0.2810	−0.3934	−0.3321	−0.3075	−0.2562	−0.2460
2	−0.0368	−0.1328	−0.1280	−0.0388	−0.0864	−0.0180	−0.0864	−0.0731
3	−0.0168	−0.0414	0.0384	0.0316	0.0552	−0.0273	0.0230	0.0176
4	0.0351	0.0948	−0.0799	−0.0472	−0.1055	0.0116	−0.0763	−0.0754
5	0.0044	−0.0178	0.1207	−0.0031	0.0566	−0.0313	0.0515	0.0537
6	−0.0561	−0.0627	−0.1022	−0.0255	−0.0229	0.0069	−0.0627	−0.0505
7	0.0218	−0.0420	−0.0346	0.0576	−0.0377	−0.0376	0.0263	−0.0091
8	−0.0056	0.0805	0.0310	0.0362	0.1083	0.1280	0.0263	0.1042
9	0.0498	0.0248	0.0161	−0.0771	−0.0682	−0.1113	−0.0275	−0.0651
10	0.0367	−0.0250	0.0614	−0.0164	−0.0568	−0.0196	0.0389	0.0109

Panel B. Cross-Correlation of Implied Volatility Changes for Calls and Puts

Lag k	Closing prices	3:00 P.M. prices	2:55–3:05 P.M. prices
−3	0.0939	−0.0144	−0.0304
−2	−0.1082	−0.0054	−0.0107
−1	0.2109	0.1178	0.0009
0	−0.1858	−0.1788	0.2738
1	0.1265	0.1572	−0.0661
2	0.0039	−0.0049	0.0424
3	−0.0044	0.0362	−0.0046

[a] $n = 252$.

6 Short summary

- The paper studies implied market volatility from S&P 100 index options.
- It uses an American-style option pricing approximation with discrete dividends.
- Closing option prices are problematic because the option market closes 15 minutes after the stock market.
- This nonsynchrony induces negative serial correlation in implied-volatility changes.
- A single option transaction is also noisy because of bid/ask bounce.
- Bid/ask bounce induces additional negative first-order serial correlation and negative call-put cross-correlation.
- Using all transactions in a 2:55–3:05 P.M. window reduces these measurement artifacts.
- Pooling calls and puts and estimating an implied index level shows that stale component-stock prices are not the main issue.
- Researchers should not attach economic meaning to implied-volatility time-series patterns unless timing, bid/ask effects, and option valuation details are carefully controlled.

7 Conclusion

7.1 Core conclusion

Harvey and Whaley show that common ways of estimating S&P 100 implied volatility mechanically generate misleading time-series patterns. Closing-price estimators and single-transaction estimators create spurious negative serial correlation and implausible negative call-put cross-correlations.

7.2 Economic intuition

Implied volatility is inferred from option prices, so every microstructure and timing error in option prices becomes a volatility error. When option and index prices are not observed simultaneously, the next day's index adjustment makes yesterday's implied-volatility error reverse. When a single transaction is used, bid/ask bounce makes volatility changes look negatively autocorrelated. Using multiple transactions around the close largely removes these artifacts.

7.3 Takeaway

The paper is a warning for empirical option research: implied volatility is not directly observed. It is an estimated object, and its dynamics can be dominated by measurement choices. Researchers should use transaction data, simultaneous index information, appropriate dividend treatment, and American-style valuation before drawing conclusions about volatility behavior or option-market efficiency.